

Rheingau

Full data report

Generated 2026-08-21

Hesse

NUTS-3 regions: DE71D

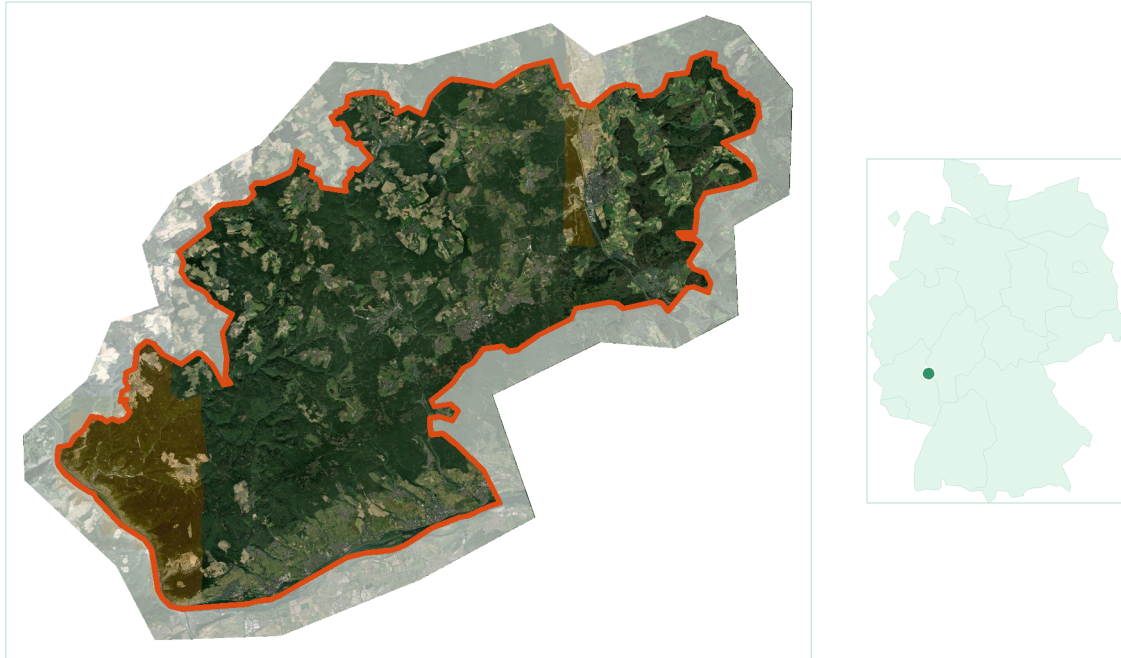


Figure 1: The Rheingau Living Lab: study region (left) and its location within Germany (right). Basemap: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Agriculture

Key figures

CROPLAND AREA	NUMBER OF FARMS	AVERAGE FARM SIZE	SHARE OF ORGANIC FARMING
11,272 ha	601 count	33.5 ha	10.9 %

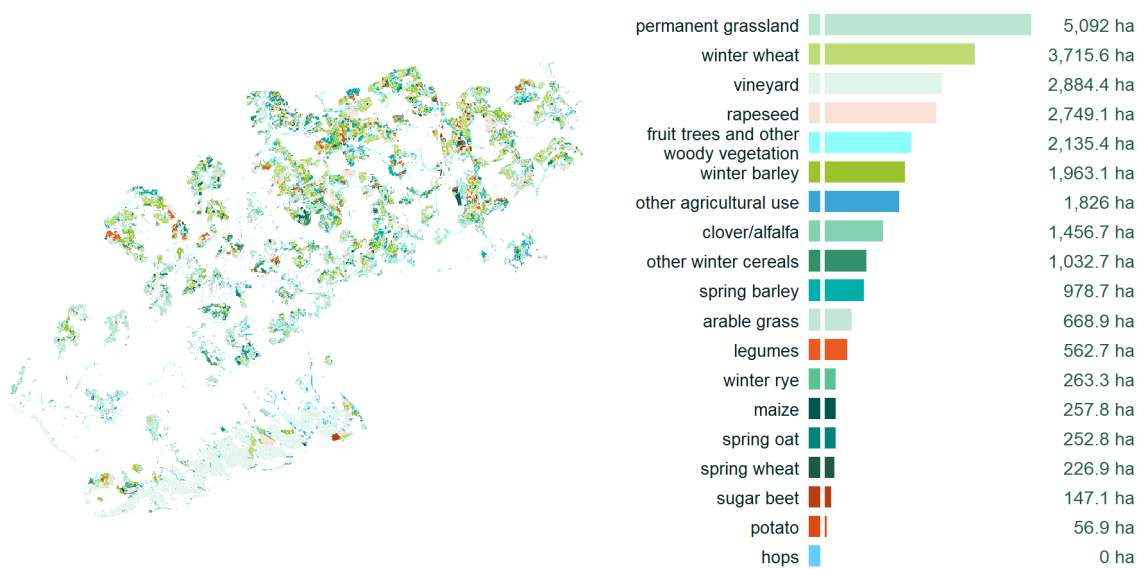


Figure 2: Map showing Agriculture in the Rheingau Living Lab (Data: Crop types (DLR, 2024))

Overview

Viticulture defines the region: around 2,700 ha of vines make up 14% of the district's agricultural area, roughly 85% Riesling and 11% Pinot Noir, worked by some 280 full-time growers — mostly small family estates alongside historic institutions such as Kloster Eberbach. Arable land accounts for 55% and permanent grassland for 31% of the roughly 19,700 ha farmed by about 620 holdings. Outside the vineyards the main branches are cereal-based arable farming, pig fattening and livery horse keeping; around 13% of farms are managed organically.

Challenges

German wine is in a structural market crisis: bulk wine prices have fallen below production costs, domestic demand for alcoholic drinks keeps declining, and trade barriers make exports harder. Steep-slope viticulture is highly labour-intensive and qualified seasonal workers are scarce, so rising wages, energy costs and regulatory burden bite hardest exactly where the landscape value is greatest. Late frost, hail, heavy rain and drought increase the risk of every vintage, and the dense clusters of Riesling remain vulnerable to fungal disease and pests. Small, fragmented holdings limit mechanisation and make succession increasingly uncertain.

Climate

Key figures

GROWING DEGREE DAYS, ANNUAL SUM (BASE 5 °C)	MEAN ANNUAL TEMPERATURE	ANNUAL PRECIPITATION	PRECIPITATION OF WARMEST QUARTER
1,834 °C·d +1,112 °C·d by 2071-2100	8.9 °C +3.9 °C by 2071-2100	726 mm +1.9 % by 2071-2100	199 mm -9.8 % by 2071-2100

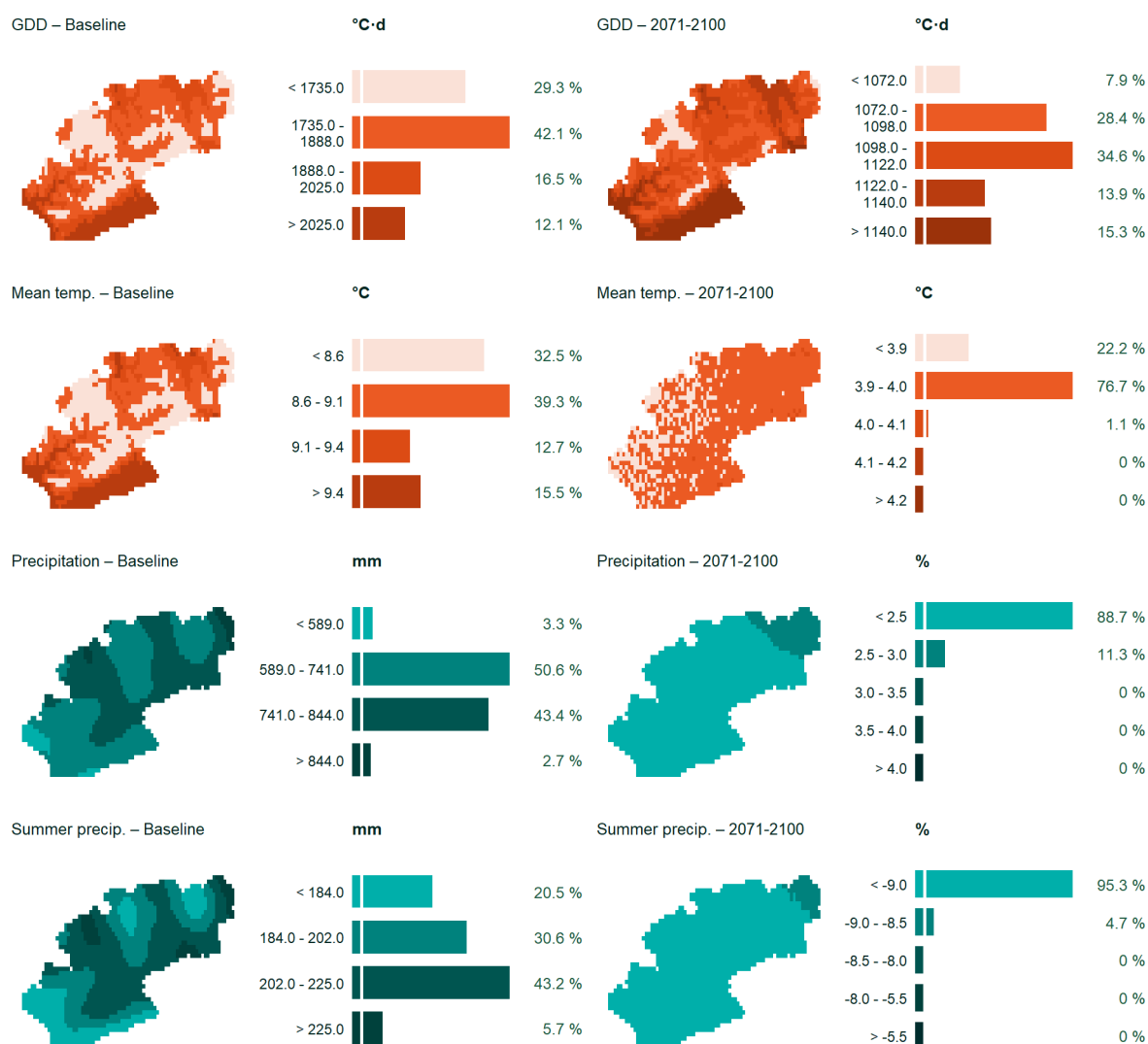


Figure 3: Map showing Climate in the Rheingau Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Heat accumulated above 5 °C over the year - a measure of how much growing season a crop gets.

Mean annual air temperature, averaged over the reference period.

Total precipitation summed over the year.

Precipitation falling in the warmest three-month quarter - relevant to summer drought stress.

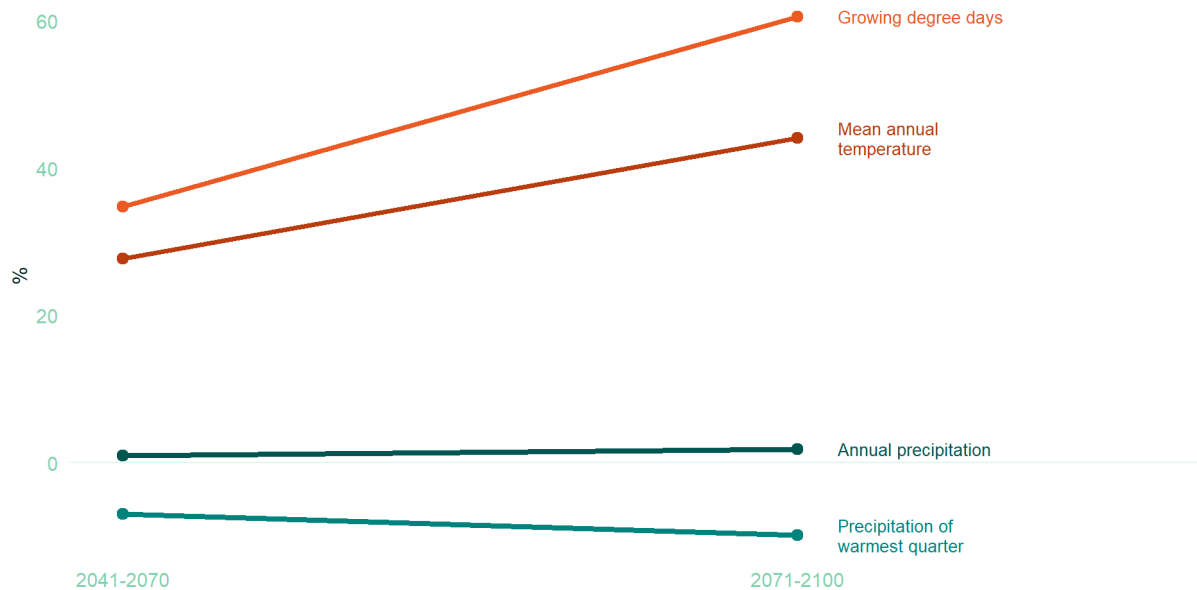


Figure 4: Chart showing Climate in the Rheingau Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Overview

Sheltered from northerly winds by the Taunus ridge and favoured by the Rhine, which stores heat and reflects light onto the south-facing slopes, the Rheingau is one of the mildest parts of Germany: an annual mean of about 10.5 °C, mild winters, warm summers and autumn mists that favour late ripening. With roughly 500—550 mm of annual precipitation it is dry by German standards. Hesse's annual mean temperature rose from 8.2 °C (1961—1990) to 9.3 °C (1991—2020), and the Rhine-Main area is among the warmest parts of the state.

Challenges

Warming is pushing Riesling towards its comfort limit: earlier budburst and ripening, higher sugar and alcohol levels and falling acidity threaten the filigree style the region is known for. Earlier budburst also raises exposure to late frost, while more hot days bring sunburn on the grapes. The climate is shifting towards an alternation of longer dry spells and heavy rainfall, so the climatic water balance deteriorates in summer and irrigation demand grows on the shallow slope soils.

Soil

Key figures

NITROGEN SURPLUS	PHOSPHORUS SURPLUS	GROUNDWATER ABSTRACTION (NON-PUBLIC SUPPLY)
No data available kg N/ha UAA	No data available kg P/ha UAA	74 1000 m3

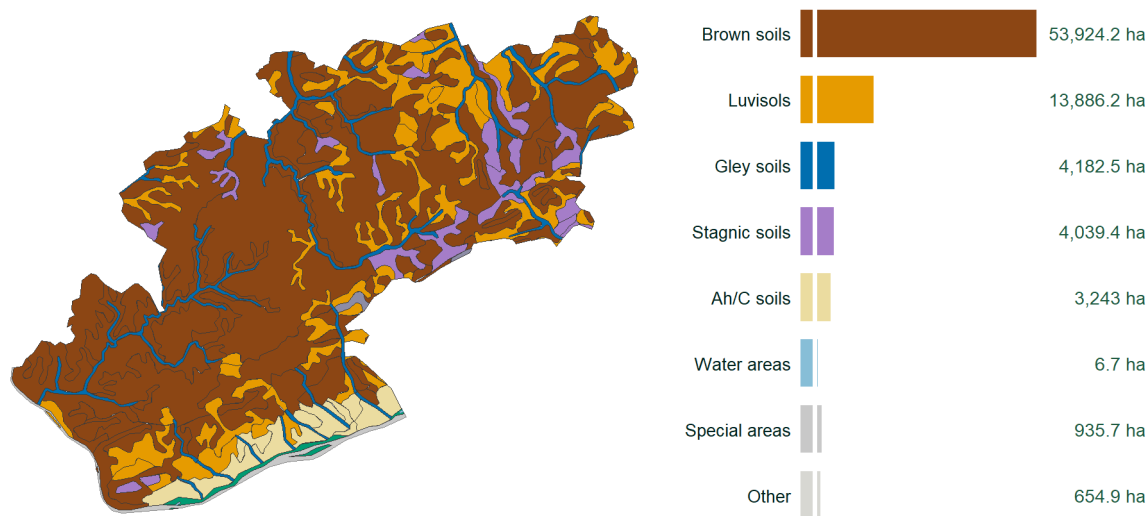


Figure 5: Map showing Soil in the Rheingau Living Lab (Data: Soil overview map (BUEK250))

Legend shows the dominant semantic soil groups for this Living Lab; raw BUEK IDs stay in the data only as provenance.

Overview

Soils change over very short distances. Deep, fertile loess and loess loam cover the gentler lower slopes and the plain near the Rhine, Taunus quartzite dominates the higher sites of the central and eastern Rheingau and the Lorch area, and phyllite and Hunsrück slate carry the vineyards of Rüdesheim, Assmannshausen, Kiedrich, Martinsthal and Rauenthal. Beneath the loess cover, weakly leached loess typically overlies deeply weathered phyllite saprolite. The stony slope soils are shallow with low water storage capacity, while the loess soils of the plain are among the most productive arable soils in the region.

Challenges

Heavy rainfall causes soil erosion on the steep slopes, particularly where rows run downslope and the ground is kept bare. Warmer soils mineralise organic matter faster, so humus is lost just where it matters most for water retention and nutrient supply. Machinery traffic compacts the soil, and nutrient leaching into groundwater remains a concern in the permanent-crop areas.

Socio-economic

Key figures

TOTAL POPULATION	GDP PER CAPITA	UNEMPLOYMENT RATE	DISPOSABLE HOUSEHOLD INCOME PER CAPITA
186,050 persons	28,672 EUR	5.1 %	31,202 EUR

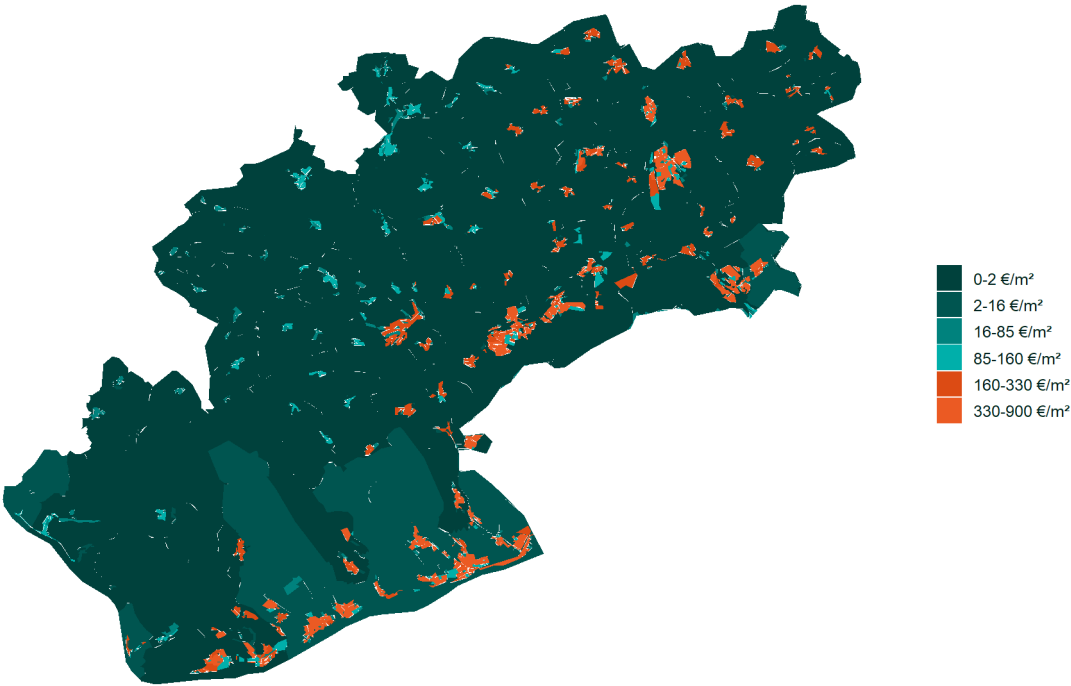


Figure 6: Map showing Socio-economic in the Rheingau Living Lab (Data: Standard land values (BORIS))

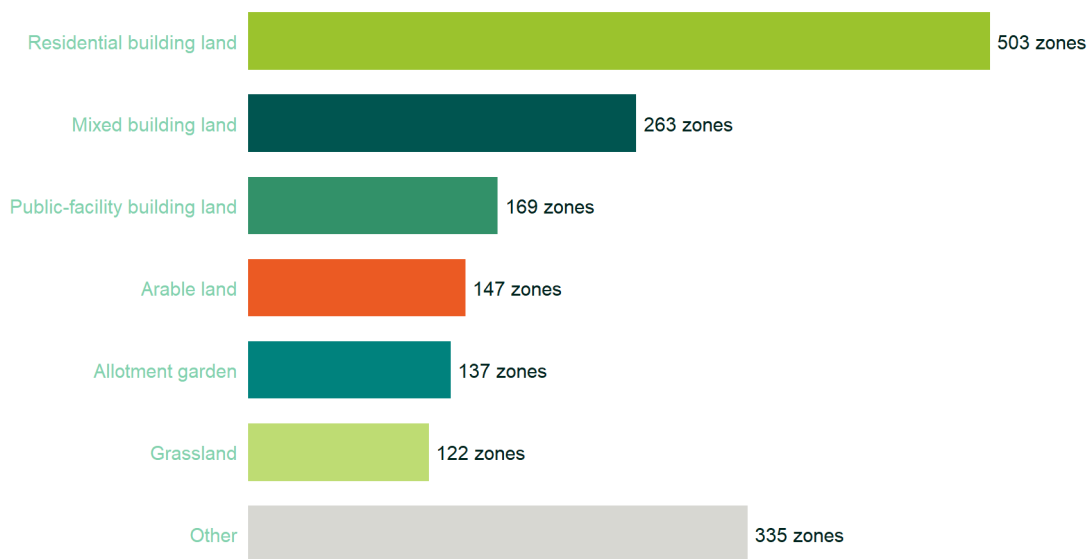


Figure 7: Chart showing Socio-economic in the Rheingau Living Lab (Data: Standard land values (BORIS))

Overview

About 185,700 people live in the district at 113 inhabitants per km², on the north-western edge of the Frankfurt/Rhine-Main metropolitan region. The economy is dominated by small and medium-sized firms, including high-tech in information and communication technology, sensor and control engineering, medical, surface and environmental technology; some 46,000 jobs and 8,000 registered businesses stand against a net outflow of commuters, mainly towards Wiesbaden, Mainz and Frankfurt. Wine and tourism are the signature sectors: around 240,000 arrivals and 701,000 overnight stays (2021) plus strong day-visitor flows, drawn by Riesling, Rhine romanticism and events such as the Rheingau Musik Festival.

Challenges

Whole villages depend economically, socially and culturally on wine: if estates disappear, the result is not only scrubbed-over slopes but empty villages, closed wine taverns and a weakened tourism offer. Falling wine prices and rising costs squeeze margins, and businesses compete for skilled labour with the far larger Rhine-Main job market, which also drives up land and housing costs. Tourism is strongly seasonal and weather-dependent, and the peripheral communities in the Taunus and the Wisper valleys are much further from centres and services than the settlements along the Rhine axis.

Landscape

Key figures

FOREST AREA	NATURA 2000 AREA	NATURE RESERVES AREA	SHARE OF SEALED SURFACE
43,842 ha	10,501.3 ha	1,221.7 ha	13.6 %

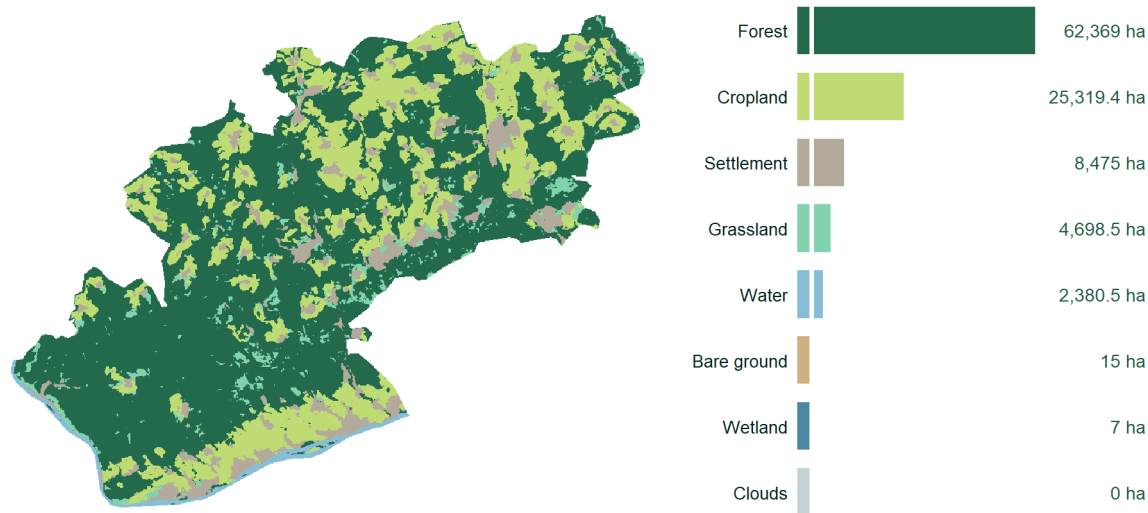


Figure 8: Map showing Landscape in the Rheingau Living Lab (Data: Land cover (Esri/Impact Observatory, 2024))

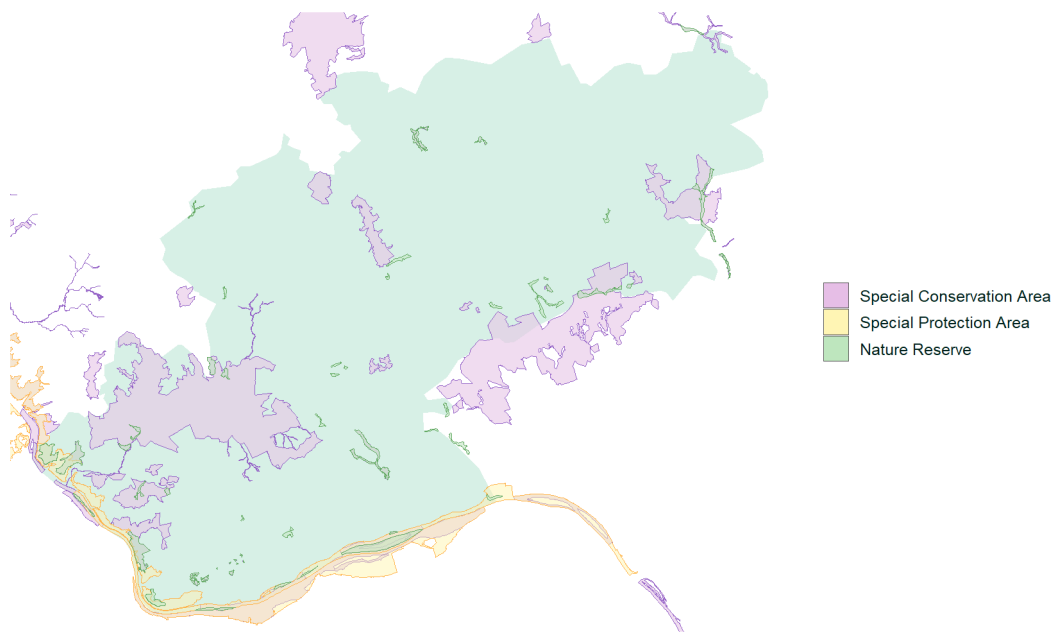


Figure 9: Map showing Protected Areas in the Rheingau Living Lab (Data: Protected areas (BfN, 2019/2023))

Conservation sites intersecting this Living Lab region. Full polygon boundaries are shown, so sites extend beyond the region outline and designations may overlap. Source: BfN (FFH and bird sanctuaries 2019, nature reserves 2023) - not suitable for planning purposes.

Overview

An east—west band between the Rhine and the Taunus, with south-facing slopes and river terraces carrying steep-slope and terraced vineyards below the wooded Rheingaugebirge. Forest covers about 56% of the district — well above the Hessian average — and the roughly 21,000 ha Hinterlandswald is the largest contiguous forest area in Hesse. The lower Rheingau from Rüdesheim to Lorchhausen belongs to the UNESCO World Heritage site 'Upper Middle Rhine Valley', with the Limes as a second World Heritage site in the district. The Rhein-Taunus Nature Park, the Wisper, Aar and Wörsbach valleys, monasteries such as Kloster Eberbach and historic wine villages complete a densely layered cultural landscape.

Challenges

Steep slopes and terraces can only be worked with high manual effort; where cultivation is abandoned, scrub encroachment erases both the terraced cultural landscape and the dry, warm habitats it carries. Heritage and scenic protection set narrow limits for new infrastructure, while land is simultaneously sought for renewable energy and settlement growth. Small, fragmented parcels make land consolidation difficult, and drought stress is increasingly visible in the forests.